

TINOTENDA MARKANDRECK MAPARA

Electronics & Computer Engineer | Software Developer | AI Researcher

Namibia

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PROFESSIONAL PROFILE

Detail-oriented Engineer with a strong background in embedded systems, circuit design, Python and C#, holding a an MSc (specialising in Artificial Intelligence) and BSc (Honours) in Electronics & Computer Engineering. Experienced in object-oriented programming, software architecture, system optimisation, and real-time applications. Proven ability to design, test, document, and maintain secure, high-performance software systems. Strong communicator with experience working in cross-functional technical teams. Highly motivated to contribute to a performance-driven engineering environment.

EDUCATION

Master of Science in Electronics & Computer Engineering

University of Namibia

Thesis: *A Multi-Sensor Fusion Framework for Micro-Drone Detection Using Acoustic and Optical Modalities*

Bachelor of Science (Honours) in Electronics & Computer Engineering

University of Namibia

TECHNICAL SKILLS

Programming Languages

- Python, C/C++, C#, JavaScript (ES6+), TypeScript
- Familiar with Rust (systems programming concepts)

Software Development

- Object-Oriented Programming (OOP)
- Software Architecture & Design Patterns

- REST APIs, Interface & Tool Development
- Code Reviews, Documentation, Testing

Data Formats & Communication

- JSON, XML, YAML
- Serial Communication, SPI, I2C, Bluetooth

Web & Backend Development

- Node.js, Express.js
- React, Angular, HTML5, CSS3
- Firebase, MySQL

Embedded & Real-Time Systems

- Arduino, ESP32, STM32, Raspberry Pi Pico, TI MSP series
- Embedded C, MicroPython/Circuit Python
- Sensor Integration, Servo Control
- Microphone Arrays, Signal Processing
- Remote Monitoring Systems

DevOps, Tools & CI/CD

- Git, GitHub, Docker
- Linux, VS Code
- Google Colab
- CI/CD Fundamentals (automated testing, version control workflows)

PROFESSIONAL EXPERIENCE

Electronics Engineer and Developer

Mwanta Digital Solutions | Namibia | 01/2024 – present

- Develop JavaScript and Python components for the Mwata system to contribute to its SAAS products.
- Design remote data collection and monitoring systems for Ongwediva Town Council.

Research & Design Intern

Sat-Com (Pty) Ltd | Windhoek, Namibia | 03/2022 – 09/2022

- Designed, programmed, and tested Bluetooth communication systems using C and embedded platforms.
- Implemented automatic pairing systems and custom device functionalities.
- Analysed and resolved signal interference affecting system performance.
- Designed a clock circuit for electronic filter optimisation.
- Actively participated in testing, documentation, and quality assurance processes.

Electronics Engineering Intern

Tula Trading CC | Ongwediva, Namibia | 05/2019 – 07/2019

- Installed and maintained CCTV, access control, and fire suppression systems.
- Analysed system faults using technical schematics and manuals.
- Ensured system compliance with design and safety specifications.
- Optimised security system performance and reliability.

SOFTWARE & SYSTEMS PROJECT EXPERIENCE

Real-Time Vehicle Tracking Web System | *Python, Node.js, WebSockets, MongoDB*

- Designed and implemented a real-time tracking platform.
- Developed REST APIs and live dashboards.
- Improved system performance and scalability through backend optimisation.

AI-Based Detection Systems | *Python, C++, TensorFlow, OpenCV*

- Designed binary and multi-class classification systems.
- Trained deep learning and traditional ML models.
- Optimised inference speed and accuracy for real-time deployment.
- Implemented automated testing and result evaluation pipelines.

Embedded Monitoring & Control Systems

- Built STM32-based monitoring systems with sensor integration and real-time communication.
- Implemented motor/servo control, remote monitoring, and data acquisition.
- Designed secure communication interfaces between hardware and software systems.

RESEARCH & PUBLICATIONS

- **Springer Nature – Journal of Intelligent & Robotic Systems (2026)**
Deep learning-based micro-drone detection using multiple sensor systems ([Deep Learning Based Micro-Drone Detection for Reliable Counter-Drone Systems using Multiple Sensors](#) | [Journal of Intelligent & Robotic Systems](#) | [Springer Nature Link](#))
- **Springer Nature – Discover Artificial Intelligence (2025)**
Lightweight machine learning models for drone detection using acoustic and optical features ([Lightweight machine learning models for drone detection using acoustic and optical features](#) | [Discover Artificial Intelligence](#) | [Springer Nature Link](#))
- **IEEE International Conference (2025)**
Performance of machine learning models in drone detection using optical features ([Performance of Machine Learning Models in Drone Detection using Optical Features](#) | [IEEE Conference Publication](#) | [IEEE Xplore](#))
- **International Engineering Conference on Sustainable Emerging Innovations and Technological Advancements (2025)**
Physical Modeling of Smart Monitoring Systems for Sustainable Water Distribution in Ongwediva Town ([Physical Modeling of Smart Monitoring Systems for Sustainable Water Distribution in Ongwediva Town](#) | [Scientific.Net](#))

LANGUAGES

- **English:** C1 – Proficient

SOFT SKILLS

- Structured, independent working style
- Strong analytical and problem-solving ability
- Team-oriented with cross-department collaboration experience
- High adaptability to new technologies

- Strong documentation and technical reporting skills
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INDUSTRIAL INTERESTS

- Industrial Software Systems
 - Embedded & Cyber-Physical Systems
 - Secure Software Architectures
 - CI/CD & Software Quality Assurance
 - Smart Infrastructure & IoT Platforms
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REFERENCES

Available upon request.